

ZIGUINCHOR, Senegal

WMO: 616950

Lat: 12.556N

Lon: 16.282W

Elev: 75

StdP: 14.66

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	62.6	64.2	36.5	31.7	80.2	39.3	35.5	80.3	16.1	78.6	13.7	80.6	3.4	0	0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	18.7	102.4	72.0	100.4	72.3	98.4	73.1	83.6	91.7	82.8	90.4	82.0	89.4	6.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.5	164.4	88.2	80.8	160.3	87.3	80.2	157.4	86.5	47.7	92.3	46.8	90.4	45.9	90.0	95.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.9	11.3	9.8	57.2	107.0	4.0	1.3	54.3	107.9	52.0	108.7	49.8	109.4	46.9	110.4	
			DB												
			WB	49.9	86.4	2.3	2.6	48.2	88.3	46.8	89.8	45.5	91.3	43.8	93.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	78.0	81.4	83.8	84.4	85.0	85.3	82.8	81.8	82.3	83.8	82.7	78.7
	DBStd	3.33	2.91	2.77	2.70	2.73	2.45	2.24	2.34	2.12	2.29	2.27	2.34	2.74
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11860	869	878	1048	1031	1084	1060	1016	987	968	1048	980	891
	CDD65	6385	404	458	583	581	619	610	551	522	518	583	530	426
	CDH74	63763	3681	4612	5988	5730	6310	6931	5565	4792	4976	6318	5213	3647
	CDH80	31121	1889	2691	3672	3433	3467	3340	2111	1596	1859	2797	2508	1758
(14) Wind	WSAvg	4.5	4.8	5.0	5.0	5.0	5.1	5.0	4.5	4.5	4.0	3.5	3.3	4.4
(15) Precipitation	PrecAvg	50.9	0.0	0.0	0.0	0.0	0.2	3.9	12.2	16.9	13.3	4.1	0.1	0.0
	PrecMax	79.1	0.3	0.2	0.0	0.0	1.1	10.4	21.6	29.4	28.0	14.2	1.6	0.6
	PrecMin	27.5	0.0	0.0	0.0	0.0	0.0	0.8	1.7	5.0	5.8	0.3	0.0	0.0
	PrecStd	11.0	0.0	0.0	0.0	0.0	0.2	2.4	4.1	6.3	4.3	3.0	0.3	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.2	102.2	105.2	105.4	103.4	99.0	94.8	92.9	94.3	96.6	96.9	97.1
		MCWB	68.0	70.1	71.3	71.8	74.3	79.1	81.0	82.4	83.0	82.7	77.5	71.4
	2%	DB	94.9	99.2	102.5	102.2	100.2	97.1	92.9	91.2	92.0	94.8	95.1	94.5
		MCWB	67.4	68.8	70.5	72.8	75.7	79.1	81.0	81.8	82.1	82.4	77.1	70.7
	5%	DB	92.9	97.1	100.2	99.6	97.2	95.2	91.2	89.6	91.0	93.2	93.4	91.8
		MCWB	67.1	68.4	70.6	73.3	76.3	79.1	80.7	80.9	81.6	81.8	76.8	70.5
	10%	DB	90.5	94.7	97.1	96.6	95.1	93.3	89.5	87.8	89.2	91.5	91.7	89.6
		MCWB	66.3	68.1	70.7	73.6	76.4	79.0	80.1	80.5	80.9	81.2	76.3	69.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.0	77.3	78.6	79.8	81.4	82.8	83.2	83.7	84.4	85.3	83.9	79.1
		MCDB	89.0	92.1	96.3	96.3	95.5	92.6	91.0	90.7	92.4	93.3	92.1	88.5
	2%	WB	72.0	74.7	76.4	77.8	79.9	81.5	82.1	82.6	83.0	83.8	81.9	76.5
		MCDB	85.7	89.9	93.5	94.5	93.4	91.8	89.8	89.1	90.4	92.0	90.1	85.1
	5%	WB	70.1	72.8	74.9	76.5	78.6	80.7	81.3	81.6	82.0	82.8	80.6	74.5
		MCDB	84.2	87.7	91.4	92.6	91.4	91.3	88.8	87.7	88.8	90.5	88.4	82.5
	10%	WB	68.4	71.0	73.3	75.2	77.5	79.9	80.5	80.7	81.2	82.0	79.3	72.7
		MCDB	82.6	85.8	89.2	90.3	89.8	89.7	87.6	86.5	87.4	89.1	86.1	81.6
(35) Mean Daily Temperature Range	5% DB	MDBR	26.3	28.6	30.3	29.1	24.9	18.7	14.1	13.0	14.4	16.5	20.4	23.6
		MCDBR	29.5	31.7	33.4	31.7	27.3	20.8	16.5	15.3	16.6	18.2	22.1	27.2
		MCWBR	11.8	11.6	10.9	9.9	8.4	6.4	6.4	7.0	7.7	7.7	8.1	10.2
	5% WB	MCDBR	27.3	29.1	30.7	29.0	23.9	18.8	15.3	14.3	15.7	17.4	18.9	23.5
		MCWBR	11.2	11.7	11.4	10.2	8.3	6.7	6.8	7.1	7.8	8.2	7.9	9.4
(40) Clear-Sky Solar Irradiance	taub		0.531	0.578	0.668	0.657	0.678	0.673	0.601	0.544	0.537	0.580	0.538	0.507
	taud		1.913	1.812	1.641	1.713	1.685	1.762	1.932	2.095	2.093	1.954	1.995	2.024
	Ebn at Noon		245	239	221	223	214	212	229	246	249	236	241	248
	Edh at Noon		60	69	84	77	78	71	61	52	52	59	55	53
(44) All-Sky Solar Radiation	RadAvg		1719	1916	2137	2220	2137	1868	1580	1506	1596	1720	1686	1631
	RadStd		87	80	69	51	82	78	83	91	80	70	73	75

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page